

AJAY KESARWANI

Senior Software Developer | AI/ML Engineer

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PROFILE SUMMARY

Senior Python Engineer with 5+ years of backend development and hands-on AI/ML engineering experience, specializing in building production-grade ML pipelines, scalable APIs, and GenAI-driven applications. Proficient in Python, FastAPI, Docker, Kubernetes, and cloud-native deployments. Experienced in performance optimization and CI/CD pipelines, with a proven ability to mentor teams, improve developer experience, and productionize ML models.

EXPERIENCE

Backend Project Developer (Freelance / Academic)

12/2023 – Present

Passau, Germany

- Designed and deployed ML/NLP models (Transformers, HuggingFace, LangChain) with FastAPI-based serving and monitoring.
- Built secure, scalable backend services with Python (FastAPI, Django, Flask), Java (Spring Boot), Hibernate, and PostgreSQL/MySQL.
- Containerized workloads with Docker and deployed on Azure Kubernetes clusters, optimizing cost and latency.
- Maintained CI/CD pipelines (GitLab, Jenkins) for automated deployment of ML models and microservices.
- Developed RAG pipelines and GenAI/LLM integrations for real-world applications.
- Provided technical documentation and mentoring to peers on Python coding best practices and ML serving workflows..

Lead Software Engineer, HCL TECHNOLOGY

12/2018 – 11/2020

Bangalore, India

- Integrated ML-driven optimization algorithms into production, improving 5G network planning efficiency.
- Developed and deployed Python-based APIs (FastAPI/Django) for ML model serving and orchestration.
- Managed containerized deployments on AWS with Jenkins-based CI/CD pipelines.
- Led root-cause analysis and performance optimization of deployed ML services.
- Mentored junior engineers on Python best practices and ML model integration.

Software Engineer, ERICSSON

02/2015 – 12/2018

Kolkata, India

- Built scalable backend systems for centralized mobile network provisioning during 4G rollouts.
- Developed and maintained SOAP/REST APIs handling millions of transactions daily.
- Automated deployments with Jenkins + Docker for distributed systems.
- Collaborated with cross-functional teams to ensure low-latency backend services.

TECHNICAL PROJECTS

-  **Master's Thesis: Exploring Constructiveness in Online News Comments**
- Built a classifier achieving 94% F1-score for detecting constructive comments.
 - Deployed a FastAPI inference API with Streamlit dashboard for real-time comment analysis.
-  **Accident Video Classification**
- The LSTM model with MobileNetV2, leveraging temporal dependencies between frames, achieved the highest classification accuracy of 82%, demonstrating applicability to autonomous driving safety systems
-  **Passenger Volume Prediction for Bus Travel**
- Implemented LSTM-based time series forecasting with AutoARIMA baseline, achieving 82% accuracy
 - Conducted data smoothing, scaling, and sliding-window optimization for time-based features
-  **Cold Email Generator using LLMs**
- Developed a LangChain-powered GenAI tool to generate personalized cold emails with semantic search
 - Integrated FAISS embeddings, OpenAI GPT-4, and ChromaDB for RAG pipeline.
 - Deployed with Streamlit + FastAPI, and added sentiment analysis and portfolio matching modules
-  **Restaurant Name and Menu Generator using LLMs**
- Built an LLM-driven tool to generate creative restaurant names and full menus based on user-selected cuisine.
 - Incorporated temperature tuning, prompt chaining, and multi-turn generation for better user control.
 - Built a tool to generate restaurant names and menus, utilizing LLMs and a temperature slider.

EDUCATION

M.Sc. in Computer Science, University of Passau

 11/2020 – 10/2024

 Germany

Thesis: *Exploring Constructiveness in Online News Comments*

Key Coursework: Computational Linguistics, Multimedia Databases, Data Science Lab

B.E. in Information Technology, IIST, Shibpur

 June 2010 – July 2014

 India

Thesis: *An Optimized Recursive Transformation Approach for Reversible Circuit Synthesis*

Key Coursework: Object-Oriented Programming, Data Structures & Algorithms, DBMS

SKILLS

- **Programming Languages & Backend :** Python, FastAPI, Django, Flask, Java (Spring Boot)
- **AI/ML:** PyTorch, TensorFlow, Scikit-learn, HuggingFace, LangChain, GenAI/LLM integration
- **Databases:** PostgreSQL, MySQL, Oracle, Redis, ChromaDB, FAISS
- **Cloud & Infrastructure:** WS, GCP, Azure, Docker, Kubernetes, Jenkins, GitHub Actions, Grafana/Prometheus
- **CI/CD & Tools:** GitLab CI/CD, SonarQube, Azure DevOps
- **Agile Tools:** Jira, Confluence, Scrum, Kanban

CERTIFICATIONS

- Machine Learning Specialization (Coursera)
- Deep Learning Specialization (Coursera)
- Oracle Certified Java Professional 6

LANGUAGES

- English (Professional)
- German (B2)
- Hindi (Native)

HOBBIES

- Chess
- Badminton
- Yoga